



AI-Powered Industrial Machine Intelligence

Navigation

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Deploy

```
<h1>  PredictX</h1>

<p>
  AI-Powered Industrial Machine Failure,
  Remaining Useful Life & Maintenance Cost Prediction
</p>
```

PredictX Overview

Machines

20

Failure Events

3,560

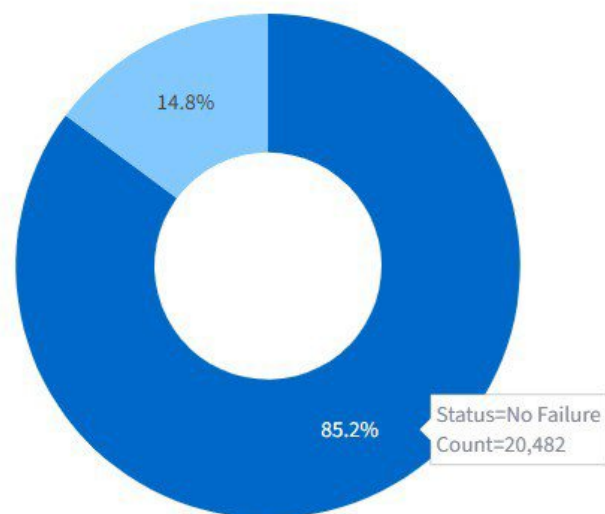
Average RUL

27.8 hrs

Avg Repair Cost

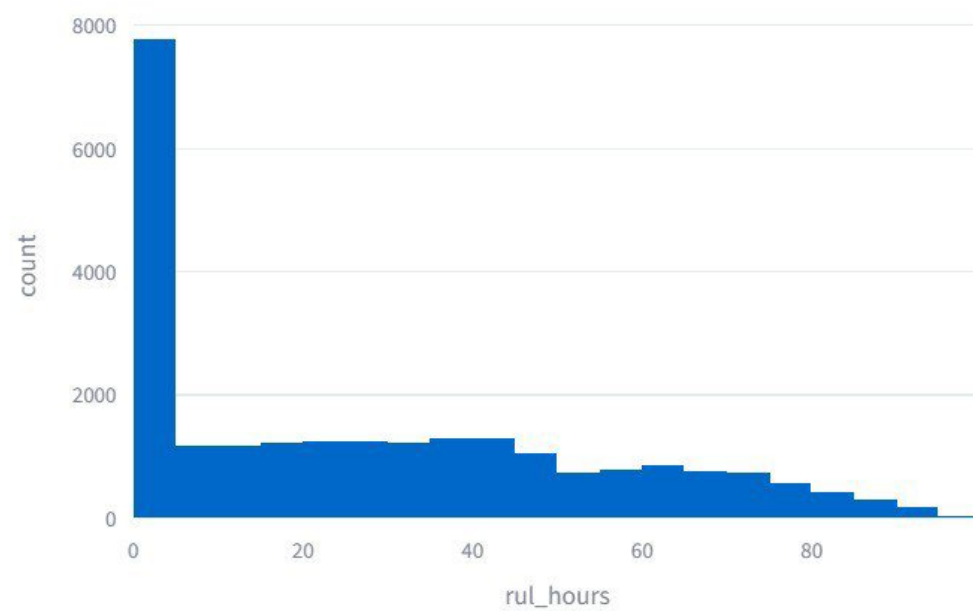
₹609

Machine Failure Distribution



Remaining Useful Life Distribution

■ No Failure
■ Failure



Machine Dataset Preview

[illegible]

Machine Dataset Preview

[illegible]

Machine Health Prediction

Enter the current machine operating conditions.

Machine ID

10

- +

Average Phase Current

6.43

- +

Hours Since Maintenance

121.61

- +

Machine Type

CNC

▼

Pressure Level

46.30

- +

Ambient Temperature

13.00

- +

Vibration RMS

1.27

- +

RPM

856.00

- +

Hour

12.00

- +

Motor Temperature

50.06

- +

Operating Mode

idle

▼

Day

15.00

- +

Day of Week

2.00

- +

Month

6.00

- +

Weekend?

0

▼

Maintenance Due

0

▼

🚀 RUN PREDICTION

🔄 Prediction started...

✅ Prediction completed successfully!

🎯 Prediction Results

✅ LOW FAILURE RISK

🕒 Remaining Useful Life

9.4 hrs

💰 Estimated Repair Cost

₹0

🔧 Predicted Failure Type

none

Failure Probability

6.08%

🕒 Maintenance Planning Recommended

The predicted remaining useful life is relatively low. Consider planning maintenance before the machine reaches a critical condition.



Machine Learning Model Comparison

Compare the performance of different machine learning models.



Comparison File

Select Prediction Task

Failure Prediction

 Comparison file loaded successfully!



Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
XGBoost	0.9738	0.9092	0.9143	0.9118	0.9954
Extra Trees	0.9713	0.8942	0.9143	0.9042	0.9945
Random Forest	0.9638	0.8431	0.9284	0.8837	0.993
Decision Tree	0.9391	0.7479	0.8876	0.8118	0.9326
SVM	0.9278	0.6868	0.9424	0.7946	0.9764



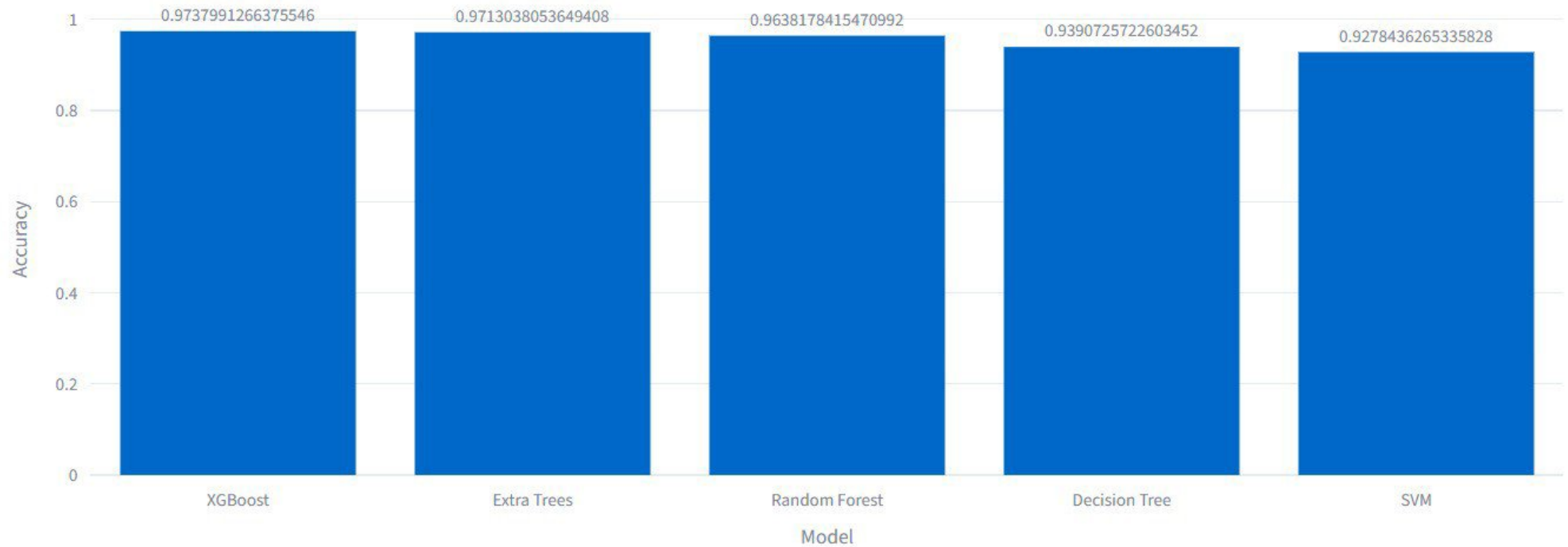
Performance Comparison

Select Metric

Accuracy



Failure Prediction - Accuracy



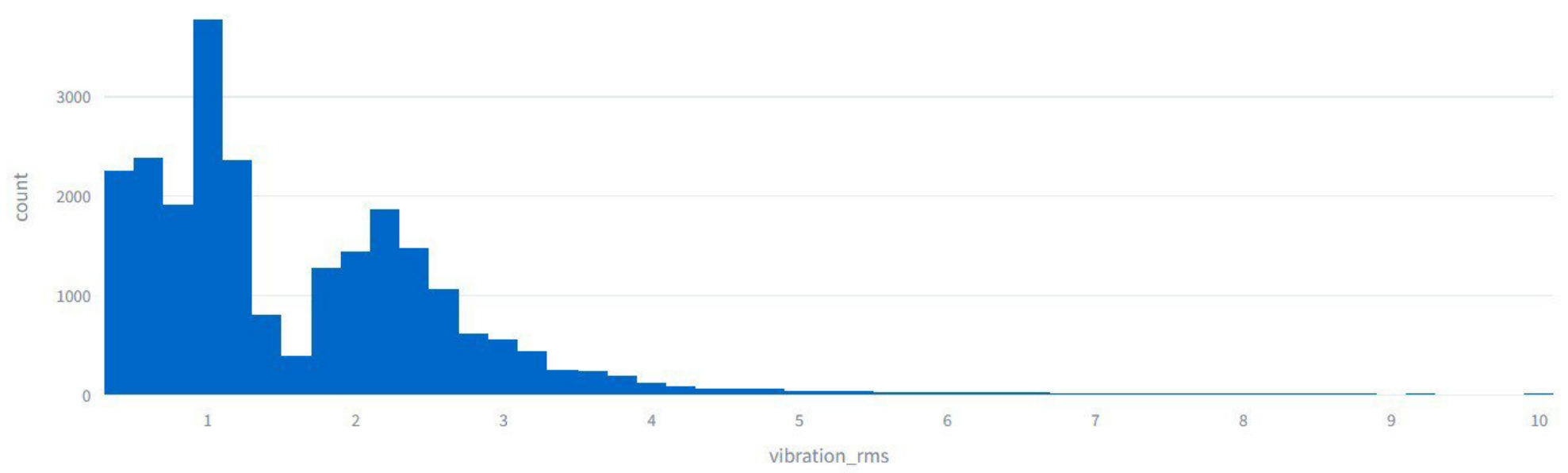
Industrial Sensor Analytics

Select Sensor / Variable

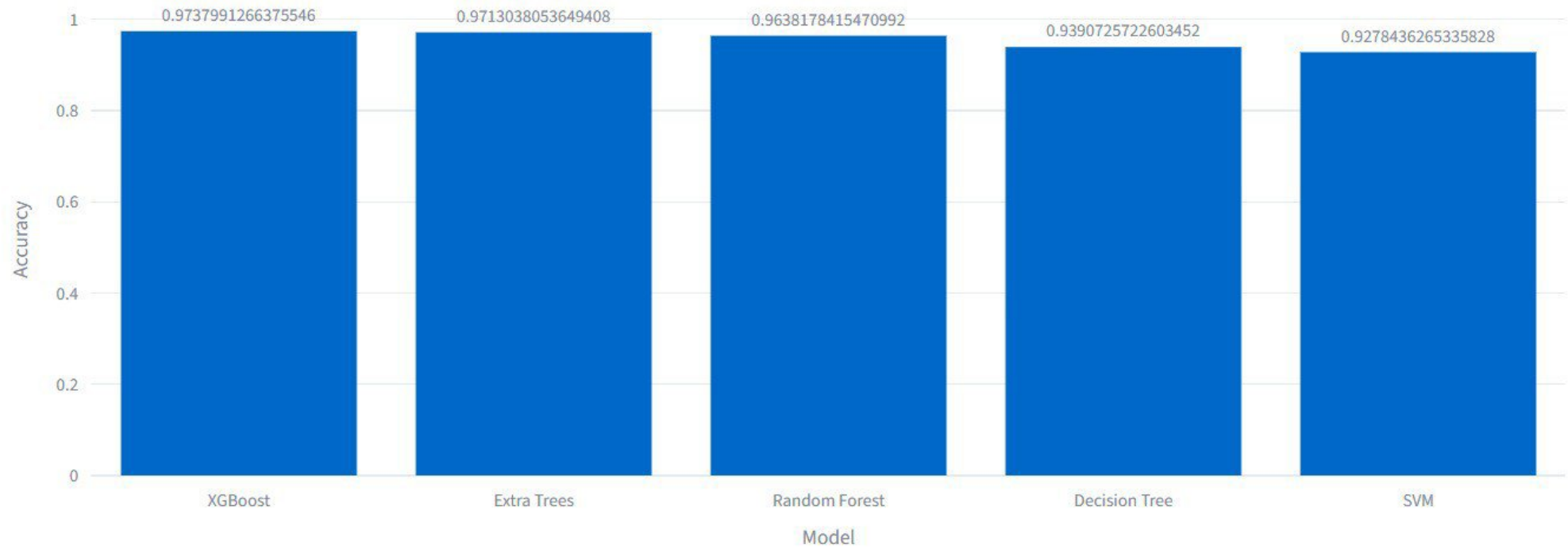
vibration_rms



vibration_rms Distribution



Failure Prediction - Accuracy



Best Model: XGBoost

Accuracy: 0.9738

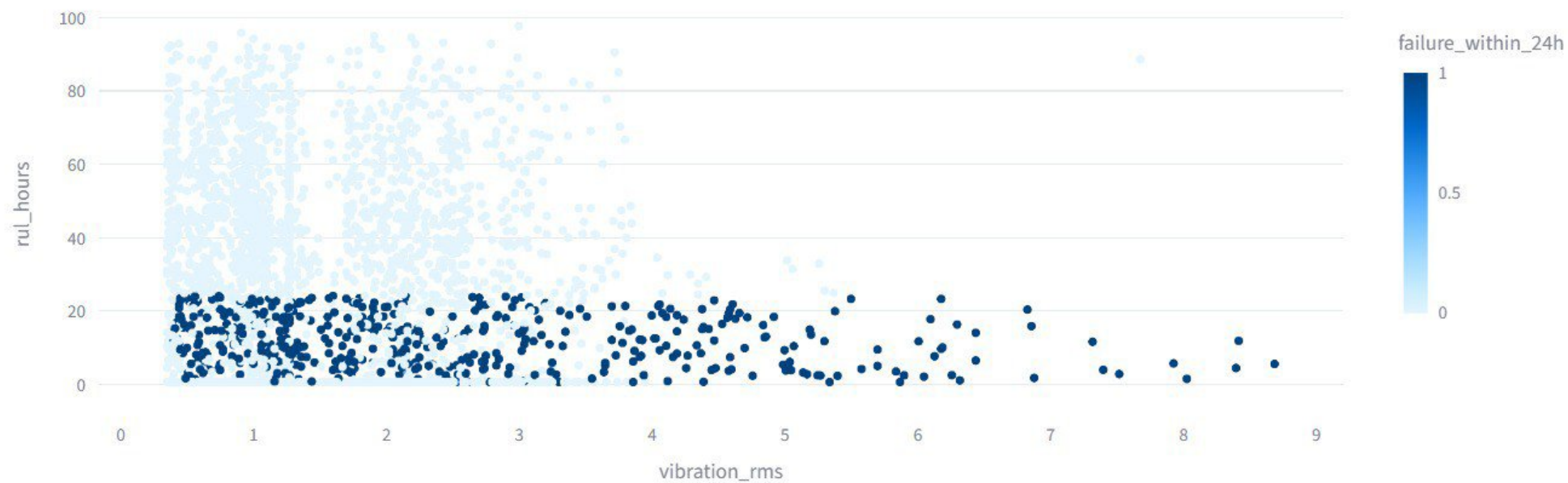
Sensor Relationship with RUL

Select Variable

vibration_rms



vibration_rms vs Remaining Useful Life



PredictX AI Insights

PredictX uses machine-learning models to support predictive maintenance decisions.

Failure Risk

Historical Failure Rate

14.81%

Machines with elevated sensor readings or long maintenance intervals may require additional monitoring.

RUL

Average Remaining Useful Life

27.8 hrs

RUL prediction helps maintenance teams estimate when intervention may be required.

Maintenance Strategy

-  Monitor abnormal vibration levels
-  Monitor motor temperature
-  Monitor phase current
-  Monitor RPM changes
-  Track hours since maintenance
-  Prioritize machines with low predicted RUL
-  Estimate repair costs before maintenance

Machine Learning

The system compares multiple machine-learning algorithms and selects high-performing models for different prediction tasks.

Failure Prediction

- Classification models

Remaining Useful Life

- Regression models
- 🏆 Extra Trees selected as the best RUL model

Failure Type

- 🏆 XGBoost selected as the best model
- Accuracy: 97.75%
- F1 Score: 97.74%

Maintenance Cost

- Regression model

Remaining Useful Life

- Regression models
- 🏆 Extra Trees selected as the best RUL model

Failure Type

- 🏆 XGBoost selected as the best model
- Accuracy: 97.75%
- F1 Score: 97.74%

Maintenance Cost

- Regression model

Dataset

Dataset records: 24,042

Dataset features: 21

Unique machines: 20

PredictX — Turning industrial data into predictive maintenance intelligence.

Explorer

PredictX

> __pycache__

> .venv

> etc

> Include

> Lib

> Scripts

> share

pyenv.cfg

data

predictx_cleaned.csv

predictx_cost_mod...

predictx_failure_m...

models

predictx_best_cost...

predictx_best_failu...

predictx_best_failu...

predictx_best_rul...

predictx_failure_pr...

predictx_failure_ty...

predictx_failure_ty...

predictx_failure_xg...

app_backup.py

app.py

requirements.txt

Outline

Timeline

Welcome × requirements.txt × app.py × app_backup.py × Release Notes: 1.135.0 ×

requirements.txt

1 streamlit

2 pandas

3 numpy

4 scikit-learn

5 xgboost

6 joblib

7 plotly

st.button

Aa ab .*

No results

↑ ↓ ≡ ×

Problems Output Debug Console Terminal Ports

(.venv) PS C:\Users\ezhilarasi.k\OneDrive\Documents\PredictX> streamlit run app.py

Help agents write better Streamlit apps?

Install the official Streamlit skills by running [streamlit skills](#) in your terminal.

2026-08-31 17:40:05.397 Please replace `use\_container\_width` with `width`.

`use\_container\_width` will be removed after 2025-12-31.

For `use\_container\_width=True`, use `width='stretch'`. For `use\_container\_width=False`, use `width='content'`.

2026-08-31 17:40:05.480 Please replace `use\_container\_width` with `width`.

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powersh... ⚠

powersh... ⚠

powersh... ⚠

powershell

streamlit

app.py

> st.button

Aa ab .* ? of 1

```
1  import os
2  import numpy as np
3  import pandas as pd
4  import joblib
5  import xgboost as xgb
6  import streamlit as st
7  import plotly.express as px
8
9
10 # =====
11 # PAGE CONFIG
12 # =====
13
14 st.set_page_config(
15     page_title="PredictX | Industrial AI",
16     page_icon="🏭",
17     layout="wide",
18     initial_sidebar_state="expanded"
19 )
20
21
```